

Formative activity

1 Context of the activity

An earthquake is the sudden release of energy accumulated within the Earth's tectonic plates. This generates vibrations that propagate in the form of seismic waves. Seismologists use scales such as the Richter scale to measure the strength of these phenomena. Table 1 shows the dates, origin and magnitude of some earthquakes that have happened in Mexico.

Date	Place	Magnitude
28 mar 1787	Oaxaca coastline	8.6
19 sep 1985	Michoacán / CDMX	8.1
7 sep 2017	Gulf of Tehuantepec	8.2
19 sep 2017	Morelos-Puebla	7.1
6 jul 1964	Tierra Caliente	7.2 – 7.3
23 ago 1965	Oaxaca	7.3 – 7.5
20 mar 2012	South of Ometepepec, Guerrero	7.5 – 7.7

Table 1: Major earthquakes in Mexico and the energy they release

The Richter scale is based on a logarithmic function that relates the amplitude of the waves to a numerical magnitude. For instance, an earthquake measuring 4.0 on the Richter scale may be perceived as a slight vibration indoors, whereas a magnitude of 6.0 could result in moderate damage to vulnerable structures. In contrast, an earthquake measuring 8.0 or higher, such as the one that struck Michoacán in 1985, can be devastating.

Using mathematical models such as the Gutenberg-Richter equation enables us to calculate the energy released during an earthquake and make quantitative comparisons between events. According to Gutenberg-Richter's law, which describes the seismic distribution of magnitude and frequency in Mexico, the typical constants adjusted for the central region of the country¹ are as follows:

$$\log [N(M)] = 4.54 - 0.55M. \quad (1)$$

Where $N(M)$ represents the number of earthquakes at the same magnitude, M , as defined on the Richter scale.

These models are essential for understanding the relationship between earthquake magnitude and the amount of energy released. For example, an increase of one unit on the Richter scale implies that approximately 32 times more energy is released. Therefore, a 7.0 magnitude earthquake

¹Equation retrieve from the journal of the Mexican geophysical union. Link to download the document

releases energy equivalent to the explosion of 500 000 tons of TNT, whereas a 5.0 magnitude earthquake releases energy comparable to 500 tons of TNT. The algebraic expression that relates the energy released by an earthquake with the following equation,

$$\log [E] = 4.8 + 1.5M. \quad (2)$$

Where M is the magnitude of the earthquake in the Richter scale and E is the energy in joules (force times distance). In this way, mathematics helps us not only to quantify natural phenomena, but also to prepare for their possible effects.

2 Activity

In this activity, we will use the given information to develop an understanding of how functions can be used to quantify natural phenomena. To do this we are going to make algebraic manipulations to the equations (1) and (2) to find the following:

- Find the algebraic expression for the difference of energy between different magnitudes.
- Calculate the energy of an earthquake in terms of the frequency of earthquakes of the same magnitude in Mexico.
- Find a relation between the energy and the number of earthquakes with the same magnitude.
- Graph the functions and analyze the order of magnitude of the energy.

In order to accomplish these objectives, fulfill the tasks below.

2.1 Exploring with algebra

First, we will find the algebraic expression for the difference in energy between earthquakes of different magnitudes. Then we are going to express this difference in terms of energy and number of earthquakes with the same magnitude.

1. Solve for $N(M)$ in equation (1)
2. Solve for E in equation (2)
3. Compute the difference $E_2 - E_1$ using the previous result. You should arrive to

$$E_2 - E_1 = 10^{4.8} \left[10^{1.5M_2} - 10^{1.5M_1} \right].$$

4. Compute the difference between $\log [E_2] - \log [E_1]$. You should get

$$\frac{E_2}{E_1} = 10^{1.5(M_2 - M_1)}.$$

5. Compute the difference $M_2 - M_1$ using equation (2). You should get

$$M_2 - M_1 = \log \left[\left(\frac{E_2}{E_1} \right)^{3/2} \right]$$

6. Compute the difference $M_2 - M_1$ using equation (1). You should get

$$M_2 - M_1 = \log \left[\left(\frac{N(M_2)}{N(M_1)} \right)^{-1/0.55} \right].$$

7. Using the two previous results, arrive at the following equation

$$E_2 = \left(\frac{N(M_2)}{N(M_1)} \right)^{-2/(3 \cdot 0.55)} E_1$$

8. Considering that 1 Ton of TNT releases 4.184×10^9 joules, use equation (2), to find the magnitude of an earthquake that releases the same amount of energy.

2.2 Exploring with graphs

Now we are going to explore the numeric values of the previous expressions.

1. Using equation (2), compute the energy of the earthquakes shown in table 1.
2. Divide the previous result by 4.184×10^9 , to express the energy in tons of TNT.
3. Graph the numbers using Excel or by hand. Use logarithmic scale in the y axis.
4. Compute the energy difference between the earthquake of march 28 and august 23. Express the energy in tons of TNT.
5. Graph the equation (2) for the earthquakes with magnitude $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$. Express the graph in terms of tons of TNT and use logarithmic scale in the y axis.
6. Identify the range and domain of equation (2) using the previous graph.

2.3 Delivery instructions

Upload a PDF document to Canvas containing the algebraic procedures. Please enclose your final answer in a box. Add the graphs and tables from the second part to the same document. Although this activity is to be completed individually, you can work in groups. Remember to include a cover page, and ensure your procedures are clear and orderly.

2.4 Algebraic tools

Finally, here are some algebraic properties that might be helpful for the activity.

Laws of exponents

$$\begin{aligned} a^r \cdot a^s &= a^{r+s}, & (a^r)^s &= a^{rs} \\ \frac{a^r}{a^s} &= a^{r-s}, & (ab)^r &= a^r b^r \\ \left(\frac{a}{b}\right)^r &= \frac{a^r}{b^r}, & b &\neq 0 \end{aligned}$$

Laws of Logarithms

$$\begin{aligned} \log_a(AB) &= \log_a(A) + \log_a(B) \\ \log_a\left(\frac{A}{B}\right) &= \log_a(A) - \log_a(B) \\ \log_a(A^C) &= C \log_a(A) \end{aligned}$$

Inverse functions property

$$(f \circ f^{-1})(x) = x = (f^{-1} \circ f)(x) \quad \rightarrow \quad \log_a(a^x) = x = a^{\log_a(x)}$$